Geo-Sentinel Web

One-Page Project Brief

Geo-Intelligence Platform

Mission: Geo-Sentinel is a full-scale AWS-hosted geopolitical intelligence website that accepts a user-defined scenario, gathers relevant global news, performs sentiment analysis across unique sources, and delivers structured, downloadable reports for basic and premium users.

Project Overview

Geo-Sentinel Web is a cloud-oriented intelligence application designed to transform raw geopolitical news into structured insights. A user enters a scenario such as *"Oil supply disruption in the Middle East"*, and the platform retrieves relevant sources, analyzes sentiment, compares narratives, and presents decision-ready results in a clean web interface.

Core User Flow

- User enters a geopolitical scenario
- System retrieves relevant global news sources
- Sentiment analysis is performed across sources
- Results are displayed in a comparison-friendly UI
- Reports are generated for download by tier

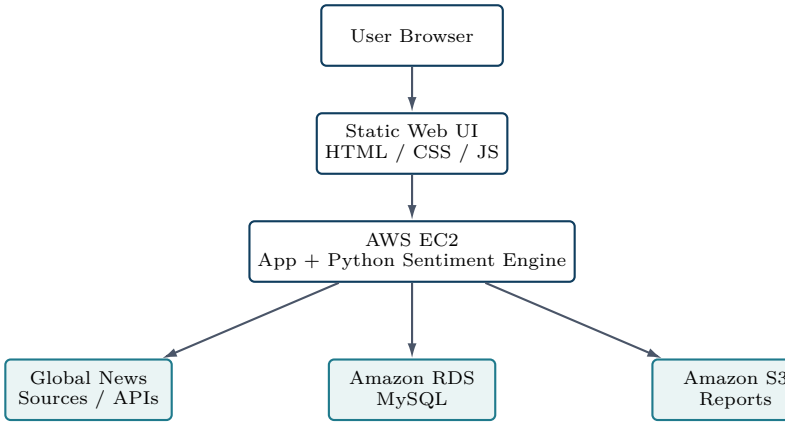
Current Implementation Strategy

- **Delivery model:** Simple-first, then scale
- **Frontend:** Static web UI using HTML, CSS, JavaScript
- **Backend path:** Python-based sentiment processing
- **Infrastructure:** AWS provisioned with Terraform
- **Near-term goal:** Finish frontend flow, then connect backend

Key Value

- Multi-source geopolitical intelligence
- Sentiment-driven comparison, not just headlines
- Structured and downloadable reports
- Useful for analysts, journalists, researchers, and decision-makers

Architecture Diagram



Technology Stack

Frontend	HTML, CSS, JavaScript
Backend	Python scripts / API layer
Compute	AWS EC2
Storage	Amazon S3
Database	Amazon RDS (MySQL)
Infra as Code	Terraform
Evolution Path	Load Balancer, Auto Scaling, later advanced cloud-native expansion

Current Status & Next Steps

- Intelligence Console UI largely established
- Scenario-driven workflow is the immediate focus
- Backend integration is the next major phase
- Real source ingestion and report generation follow

Immediate next steps

- Complete scenario engine flow
- Connect UI to Python backend/API
- Integrate real news source retrieval
- Persist results and generated reports

Geo-Sentinel Web is being developed as both a client-ready geopolitical intelligence platform and a practical learning vehicle for full-stack development, AWS, Terraform, and DevOps.